Lab 7 – Prompts 01 & 02: Research Workspace and Evidence-Based Gap Matrix

Oct 22, 2026 · @Vibhum Sharma

This submission covers Phase I, Prompts 01 and 02 only, as instructed on Google Classroom. Prompts 03–18 come in later weeks.

Item	Detail
Course	AI Tools for Research — post-mid-sem assignment (Lab 7)
Student	Vibhum Sharma (Topic ID T32)
Review being converted	*Orchestrating the Virtual Tumor Board: A Comparative Literature Review of AI- and Multi-Agent-Assisted Oncology Decision Support* (Lab 4, 24 papers)
AI tool used	Claude (claude.ai, project workspace with the review PDF and 7 original papers attached)
Repository	AITR_Lab02_Project (local git repo; GitHub push pending)

Evidence labels used throughout: VERIFIED = checked in the original source or by running code. **INFERRED** = plausible, still needs testing. **UNKNOWN** = no evidence yet.

Prompt 01 — Set up an accountable research workspace

Prompt as used (only [TOPIC] was replaced):

> I am a postgraduate student converting my completed literature review into an implementation-based research paper. My topic is **Orchestrating the Virtual Tumor Board: AI- and multi-agent-assisted oncology decision support**. Act as a critical research mentor and technical pair programmer. Guide me in stages: evidence-based gap identification, literature-to-experiment mapping, feasibility, research questions, verified baseline, coding, experimental evaluation, manuscript and reproducibility. Mark statements as VERIFIED, INFERRED or UNKNOWN, and explicitly request source files or outputs when evidence is missing. Never invent papers, datasets, repository links, code execution, experiments or metrics. Help me create a portable research_decisions.md log containing decisions, reasons, source/evidence and next actions. Start by asking me to attach my existing review and summarize its scope without assuming unseen content.

What happened: the AI asked for the review. I supplied `LiteratureReview_T32_VibhumSharma.pdf` (17 pages), plus the original PDFs already in my course project. The AI extracted the full text (6,459 words) before summarising anything. Outputs A and B follow.

Output A — Workspace and decision log

The workspace is a git repository that follows the structure in Section 9 of the assignment. It was created and first committed on 20 Oct 2026 (VERIFIED: `git log`).

```
AITR_Lab02_Project/
  README.md                  research question + reproducibility
  research_decisions.md      decisions → reasons → evidence → next action
  ai_assistance_log.md       prompts, AI proposals, human checks
  docs/phase1/
    prompt01_review_scope.md
    prompt02_gap_matrix.md
```

research_decisions.md — entries made during Prompts 01–02

ID	Date	Decision	Reason	Evidence (label)	Next action
L-01	20 Oct 2026	Stay in the virtual-tumor-board area and convert the Lab 4 review.	The continuity rule in Section 2 of the assignment.	Assignment PDF (VERIFIED)	Summarise review scope
L-02	20 Oct 2026	Use the Lab 4 review (24 papers) as the evidence base, not the AI-generated Lab 1 paper.	Lab 1 was an audit of AI-written citations; Lab 4 is my own reviewed set.	Review PDF text extracted (VERIFIED)	Extract gaps

ID	Date	Decision	Reason	Evidence (label)	Next action
L-03	22 Oct 2026	Treat the review's own cross-references as unreliable.	The text cites "Section 9", "6.4" and "5.2", but the review is numbered 1–9.	Review text (VERIFIED)	Fix these before reusing text in paper 2
L-04	22 Oct 2026	Rank gaps on significance, resources and feasibility.	Free Kaggle/Colab GPUs and a GTX 1050 Ti, about 4 weeks, no access to patient data.	Own constraints (VERIFIED)	Score five gaps
L-05	22 Oct 2026	Provisionally select Gap 4 (independent, call-matched test of multi-agent boards).	It ranked first (see Output C).	Gap matrix below (INFERRED until novelty check)	Prompt 03/04: novelty search on MDAgents and CoMMa tables

Output B — Scope of the attached review

The review is a narrative synthesis of 24 papers (2011–2026) on AI support for tumor boards. It has no data or experiments of its own. All statements below are VERIFIED against the extracted review text; none of them checks the underlying papers.

Aspect	What the review covers
Question	How AI has been used and evaluated as decision support for multidisciplinary and virtual tumor boards
Corpus	24 papers found with Semantic Scholar, ResearchRabbit, Litmaps and Elicit; seeds carried over from Labs 1–3
Three technology generations	Rule-based CDSS (Watson for Oncology, 2018–2021) → single-pass LLMs (2023–2026) → multi-agent systems (2025–2026)
Five themes	AI–MDT concordance; multi-agent architectures; VTB implementation; systematic reviews and meta-analyses; cancer-type and task-specific studies
Headline claims	Concordance "typically 60% to over 90%" (Abstract); 23–99% across 59 studies (Wu et al. 2026); EvoMDT "comparable to human MDTs", 30–40% faster (Liu et al. 2026)

Aspect	What the review covers
Central pattern	The "complexity gap": AI does well on guideline-clear cases and worse on rare, multimodal or genomically complex ones
Stated gaps (Sec. 7)	Few prospective or blinded studies; no patient-outcome links; no reporting standard; thin independent validation of multi-agent systems; LMIC settings missing; cost and governance not studied

What the review does not cover:

- No benchmark datasets, no code and no released models are analysed. Only EvoMDT's public QA benchmarks are mentioned.
- Imaging, pathology and genomic inputs come up only as a limitation, not as a theme.
- No AI study from a low- or middle-income setting. India appears only through the non-AI National Cancer Grid VTB.

Housekeeping problems found while summarising (VERIFIED):

- Doğan et al. (2025), Konzmann et al. (2026) and D. Wang et al. (2025, *iScience*) are in the reference list but are never cited in the text.
- Lukac et al. (2023) is cited in Section 2.2 but is not in the reference list.
- "Wang et al. (2025)" in Section 3.2 is ambiguous, because two Wang 2025 papers are listed.
- Internal section cross-references do not match the numbering (decision L-03).

Prompt 02 — Extract evidence-based research gaps

Prompt as used ([ATTACH PAPER] = the same review PDF):

> Analyze my attached literature review **[LiteratureReview_T32_VibhumSharma.pdf]**. Identify central methods, datasets, measured findings, author-reported limitations and unresolved questions. Propose five computationally testable research gaps. For each, cite the specific passage or reference in my review; clearly distinguish a limitation reported in the original paper from your own inference; state the experimental question it suggests and what evidence would test it. Flag any claims needing verification against the original publications. Rank three gaps by research significance, available resources and implementation feasibility. Do not claim novelty without a separate literature search.

How the limitation sources were checked: the AI read the review, then searched the 7 original papers available in my project for each limitation. A limitation is marked "original paper" only where that paper's own text states it; everything else is labelled as the review's synthesis or my inference.

Output A — What the review contains

The literature mostly measures agreement with past tumor-board decisions on small, single-centre case series. Only one multi-agent system (EvoMDT) reports benchmark numbers.

Element	Extracted from review (section)	Checked in original?
Methods	Retrospective concordance (% agreement, Cohen's kappa) (3.1). Single-pass LLM prompting with a case summary, sometimes with injected guidelines (2.3, 3.1). Multi-agent systems with retrieval and consensus (2.3, 3.2). Systematic reviews and meta-analysis (3.4).	—
Datasets	Institutional case series, "commonly 10–100 cases" (6). EvoMDT: six public QA sets and four real-world sets (3.2).	EvoMDT: MedQA, MedQA-Cancer, MedMCQA-Cancer, MedQuAD, MedQARo, ROND; real-world data "on request" (VERIFIED)
Measured findings	WFO 93% (breast, 638 cases); 23–99% across 59 studies; EvoMDT beats Llama-3-70B, Claude-3 and Med-PaLM 2 and is 30–40% faster (2.2, 3.2)	EvoMDT MedQA 0.86 vs 0.76 (Llama3-70B) and 0.79 (Claude-3); 30–40% faster (VERIFIED). Park: 86.96% WFO–MTB, 322 gastric patients (VERIFIED; not quoted in the review)

Element	Extracted from review (section)	Checked in original?
Author-reported limitations	Retrospective, single-centre, small samples; no outcome link; inconsistent metrics (6)	Park: retrospective single-centre; concordance only; no survival, time or cost data; guidelines differ by country (VERIFIED). EvoMDT: limited scale and geographic diversity; no prospective or outcome study; GI non-HCC, gynaecologic, sarcoma and rare tumours untested (VERIFIED). Thiagarajan: no patient follow-up data (VERIFIED). X. Wang (iScience): data bias, uneven generalizability, limited prospective validation (VERIFIED)
Unresolved questions	Do AI-assisted boards change outcomes? Do multi-agent gains hold independently? Do results transfer to LMIC settings? Can text-only systems handle multimodal cases? (7, 8)	—

Output B — Gap matrix: source evidence → candidate question → possible experiment

All five gaps can be tested on public data with open models on free GPUs. None needs patient records.

Gap	Review passage	Original paper says vs. my inference	Experimental question	Evidence that would test it
G1. Complexity gap	Sec. 6 "complexity gap"; Sec. 3.1 Seshachalam "AI paradox"	**Original:** Park — concordance falls with age, performance status and stage IV (VERIFIED). **Inference:** that exam vignettes can stand in for case complexity.	Does LLM accuracy on oncology vignettes drop as a pre-defined complexity proxy rises, and does a multi-agent board narrow that drop?	Accuracy with 95% CI per stratum (e.g., USMLE Step 1 vs Step 2/3, or model agreement); stratum × method interaction
G2. Answer instability	Sec. 2.2 "inconsistent recommendations across repeated queries"; Sec. 6	**Review synthesis only.** Its source (Lukac et al.) is missing from the reference list, and no available original states it (UNKNOWN).	How often does the same model give a different answer to the same oncology case under resampling, paraphrase or option reordering?	Answer-flip rate and Fleiss' kappa over 10 runs per item; accuracy change under option shuffling

Gap	Review passage	Original paper says vs. my inference	Experimental question	Evidence that would test it
G3. Transfer across countries and guidelines	Sec. 7 fifth gap (LMIC settings)	**Original:** Park — Korean guidelines differ from the US-based WFO knowledge (VERIFIED); EvoMDT — limited geographic diversity (VERIFIED). **Inference:** that exam source is a usable proxy for practice setting.	Does a model's oncology accuracy differ between US (MedQA) and Indian (MedMCQA) exam items once topic is matched?	Accuracy gap with CI on topic-matched subsets; error categories where guidelines differ

Gap	Review passage	Original paper says vs. my inference	Experimental question	Evidence that would test it
G4. Multi-agent claims lack independent, fair comparison	Sec. 7 fourth gap; Sec. 3.2 recency; Sec. 6 "new failure surfaces … not yet well characterized"	**Original:** EvoMDT needs external validation (VERIFIED); its backbone is DeepSeek V3/R1, but its MCQ baselines are other single-pass models (VERIFIED). **Inference:** the gain may come from more inference calls or a different backbone, not from deliberation. No same-backbone, call-matched baseline found in the text (INFERRED; figures not yet checked).	With the backbone fixed, does a specialist-role board beat the same model voting over the same number of calls? Do roles matter? Does discussion spread errors?	Paired accuracy (McNemar test), board vs self-consistency@9; role ablation; correct→wrong flip counts; tokens per item

Gap	Review passage	Original paper says vs. my inference	Experimental question	Evidence that would test it
G5. Text-only inputs	Sec. 8 fourth direction; Sec. 2.2 "cannot access imaging or pathology data"	**Original:** EvoMDT — untested on raw unstructured records (VERIFIED). **Inference:** that adding image data changes accuracy; no reviewed study tests this.	Does adding an image (or its report) to the text case change a vision-language model's oncology answers?	Text-only vs text+image accuracy on a public multimodal oncology benchmark (access UNKNOWN)

Considered and dropped: the heterogeneity of concordance metrics (Sec. 6, Carl 2024, Wu 2026) and the missing link to patient outcomes (Sec. 7). Both need case-level tumor-board decisions or follow-up data, and no public source exists, so they are not computationally testable here.

Output C — Ranking of the top three gaps

G4 ranks first and is my provisional choice. G1 ranks second and can run as a secondary analysis inside the same G4 experiment, at no extra compute. Scores run from 1 (weak) to 3 (strong).

Rank	Gap	Significance	Resources	Feasibility (4 weeks, free T4 GPUs)	Total
1	G4 Multi-agent fair comparison	3 — tests the review's strongest claim, which also has the least validation	3 — MedQA (MIT licence, VERIFIED); open 7B model	3 — text-only, inference-only	**9**
2	G1 Complexity gap	3 — the review's central cross-cutting pattern	3 — same data	2 — a complexity proxy for exam items is weak and must be justified	**8**
3	G2 Answer instability	2 — a known LLM issue, less specific to tumor boards	3 — same data	3 — resampling only	**8**

Rank	Gap	Significance	Resources	Feasibility (4 weeks, free T4 GPUs)	Total
—	G3 Country transfer	2	3 — MedMCQA (MIT licence, VERIFIED)	2 — exam style is confounded with guidelines	7
—	G5 Multimodal	3 — fits my PhD focus	1 — dataset access UNKNOWN	1 — VLM inference on T4 is tight	5

G1 and G2 tie on total. G1 ranks higher because it is the pattern the review's conclusion rests on.

No novelty is claimed. Whether MDAgents (NeurIPS 2024), CoMMa (2026) or other medical multi-agent studies already include a same-backbone, call-matched baseline is **UNKNOWN**. That check is the first task of Prompts 03–04.

Evidence ledger — review claims checked against originals

Five claims hold up, four need correction before reuse, and four cannot be checked until I get the originals.

Claim in my review	Status	What the original shows / action
EvoMDT beats Llama-3-70B, Claude-3 and Med-PaLM 2 (3.2)	VERIFIED, with caveat	MedQA 0.86 vs 0.76 / 0.79. Mostly true, but on MedQuAD EvoMDT ties Med-PaLM 2 (BERTScore 0.63 vs 0.63), so "outperforms" should be "matches or outperforms".
EvoMDT is comparable to human MDTs and 30–40% faster (3.2, Table 1)	VERIFIED	Stated in the EvoMDT abstract; speed measured on structured dossiers only
EvoMDT needs external validation (3.2)	VERIFIED	The authors' own limitations section
Park & Chae: WFO–MTB concordance in gastric cancer (3.5)	VERIFIED	86.96%, 322 patients; published online 2023, issue 2024, 55:365–372. Add the figure to the review.
Thiagarajan: national VTB programme in India (2.1)	VERIFIED	Also reports no patient follow-up data. Supports the outcome gap.

Claim in my review	Status	What the original shows / action
EvoMDT used "six public **oncology** QA datasets" (3.2)	NEEDS CORRECTION	EvoMDT calls MedQA "a general medical examination dataset". Reword to "medical and oncology QA datasets".
"Wang, D., et al. (2025)", *iScience* (references)	NEEDS CORRECTION	The first author is Xiaodong Wang
"Lukac et al., 2023" (2.2)	NEEDS CORRECTION	Cited in the text but missing from the reference list. Find and verify it, or remove it.
Concordance "60% to over 90%" (Abstract) vs "~56–87%" (Table 1) vs 23–99% (Wu)	NEEDS CORRECTION	The review is internally inconsistent; state the range with its source
WFO 93%, 638 breast cases (Somashekhar 2018)	UNKNOWN	Original not in my files
23–99% across 59 studies (Wu 2026)	UNKNOWN	Original not in my files
"AI paradox" (Seshachalam 2026)	UNKNOWN	Conference abstract; obtain the full text
Carl 2024: no LLM-specific reporting standard	UNKNOWN	Original not in my files

Extra source found: Zengin Deniz et al. (2026, *Updates in Surgery*) is in my project files but not in the review. In 47 general-surgery patients, 38.3% of ChatGPT recommendations were fully concordant and 85.1% at least partly concordant (VERIFIED). Discordance centred on resectability and staging, which supports G1.

AI-use record for Prompts 01–02

Claude did the text extraction, the passage matching and the first draft of every table. Every number above was checked against a source text, either the review or an original PDF.

Date	What the AI did	How it was checked	Outcome
20 Oct	Proposed G4 from general field knowledge before reading the review	Re-derived from review Sec. 7 and the EvoMDT PDF	Kept; now grounded in the review's own fourth gap

Date	What the AI did	How it was checked	Outcome
20 Oct	Looked up venues for MedAgents, MDAgents and Wang et al. 2024	ACL Anthology and NeurIPS pages opened	Venues VERIFIED; their results not yet read
22 Oct	Extracted the review and 7 originals; matched limitations to sources	Quoted lines located in each PDF's text	Found four review errors (ledger)
22 Oct	Wrote that EvoMDT "outperforms" all baselines	Re-read the EvoMDT results text	Corrected: ties Med-PaLM 2 on MedQuAD

Limits of this check: text was extracted from the PDFs, so figure contents were not read. Four cited claims are UNKNOWN because I don't have their originals.

Next steps (Prompts 03–04, upcoming weeks)

- [ ] Confirm or change the G4 choice and the ranking weights myself
- [ ] Novelty search: check whether MDAgents, CoMMa and MedAgents report same-backbone, call-matched baselines
- [ ] Read EvoMDT Figure 2 for any DeepSeek-only baseline
- [ ] Obtain the Somashekhar, Wu, Carl and Seshachalam originals
- [ ] Fix the four NEEDS CORRECTION items in the review